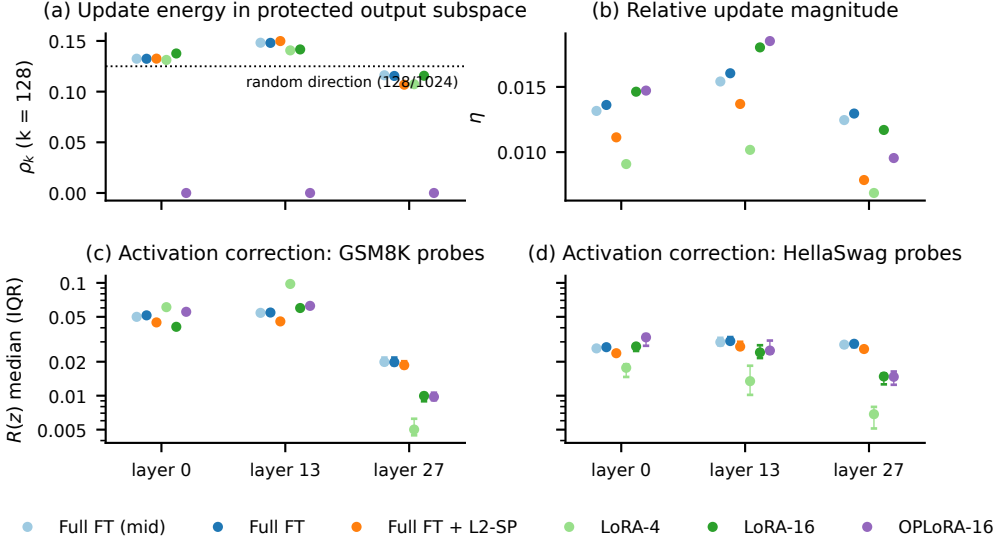
(a) Update energy in protected output subspace
ρ_k (k = 128)
0.15
0.10
0.05
0.00
random direction (128/1024)
(b) Relative update magnitude
η
0.015
0.010
(c) Activation correction: GSM8K probes
$R(z)$ median (IQR)
0.1
0.05
0.02
0.01
0.005
layer 0
layer 13
layer 27
(d) Activation correction: HellaSwag probes
layer 0
layer 13
layer 27
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16